

MATH 3310 HOMEWORK ASSIGNMENT 4

DUE ON THURSDAY 22 SEPTEMBER 2022

Please turn in your homework in class or to Weeks Hall 347 no later than 5 pm.

- (1) Construct open sentences $p(x)$ and $q(x)$ over the domain $S = \mathcal{P}(\{\triangle, *\})$ such that the all of the following hold:
- For some $x \in S$ both statements $p(x)$ and $q(x)$ are true
 - For some $x \in S$ both statements $p(x)$ and $q(x)$ are false.
 - For some $x \in S$ the statement $p(x)$ is true and $q(x)$ is false.
 - For some $x \in S$ the statement $p(x)$ is false and $q(x)$ is true.

- (2) Consider the sets

$$X = \{x \in \mathbb{R} \mid |x| \leq 3\} \quad \text{and} \quad Y = \{y \in \mathbb{R} \mid |y| = 2\}.$$

Draw the subset $(X \times Y) \cup (Y \times X)$ of $\mathbb{R} \times \mathbb{R}$.

- (3) Given the sets

$$A = \{1, 2, 4, 7, 11\} \quad \text{and} \quad B = \{1, 3, 4, 9\}$$

determine the truth value of the following statements:

- (a) $(|A| = 5) \wedge (|B| = 4)$.
- (b) $(|A - B| = |A \cap B|) \vee (|A \cup B| = 9)$.
- (c) $(A \subseteq B) \implies (A \cap B = \emptyset)$.

- (4) Construct a truth table to decide if the statement

$$(p \implies q) \wedge \sim q \implies \sim p$$

is a tautology.

- (5) Set $S = \{2, 3, 4, 5, 6, 7, 8, 9, 10, 11\}$ and consider the open sentences

$$p(x): x \text{ is prime, } x \in S \quad \text{and} \quad q(x): 2^x - 1 \text{ is prime, } x \in S.$$

- (a) Determine all values of the variable $x \in S$ for which the statement $p(x) \implies q(x)$ is true.
- (b) Determine all values of the variable $x \in S$ for which the statement $q(x) \implies p(x)$ is true.
- (c) Determine all values of the variable $x \in S$ for which the statement $p(x) \iff q(x)$ is true.